

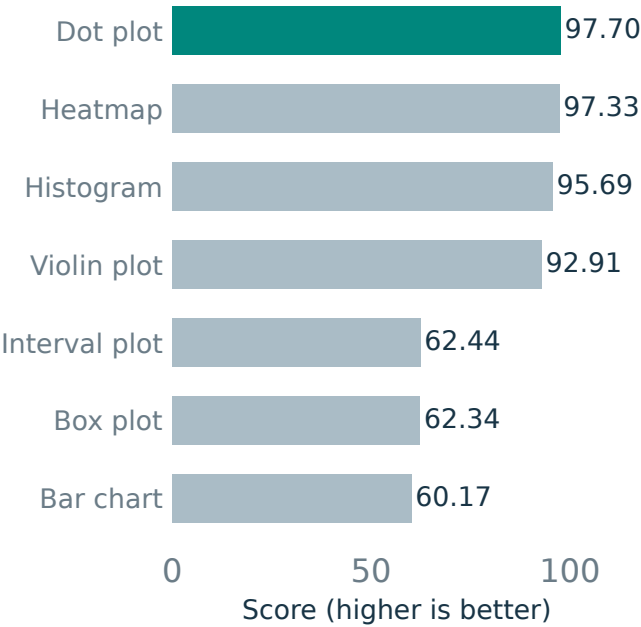
Visible differences. A traceable score.

Example: cell-nucleus area in benign and malignant samples

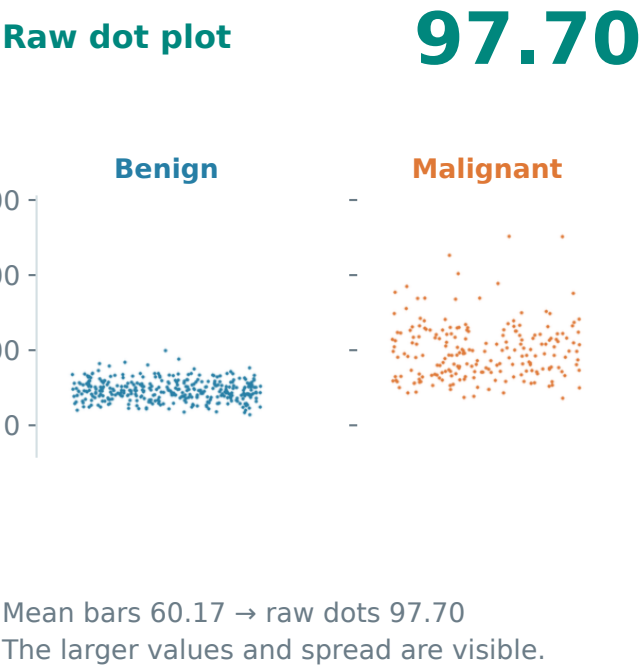
QUESTION

How does the group difference change from low to high values?

a Optimize each familiar family



b Inspect the best chart



c Read five gaps and calculate the score

Reading error = |gap recovered from chart – true gap|

Percentile	10% Low	25%	50% Middle	75%	90% High
True group gap	284.270	327.100	473.600	652.650	737.040
Mean absolute reading error	2.0302	2.5771	9.2397	5.6249	9.7958

Average error 5.85355 ÷ data spread 248.3973 = 0.023565

Score = 100 / (1 + 0.023565) =

97.70

100 means exact recovery of the requested gaps, not 100% correct readers.

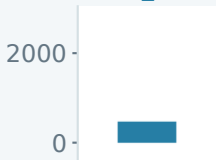
a Cell-nucleus area

How do five group gaps differ?

1 Benign

2 Malignant

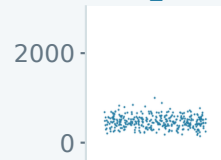
Bars 60.17
1



2



Dots 97.70
1



2



Area (source units)

+37.53 points | best of 7 familiar charts

b Age in the Cleveland cohort

How do five group gaps differ?

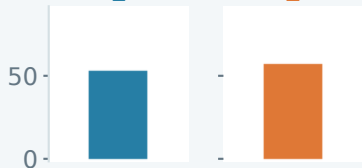
1 No disease

2 Disease

Bars 81.04

1

2

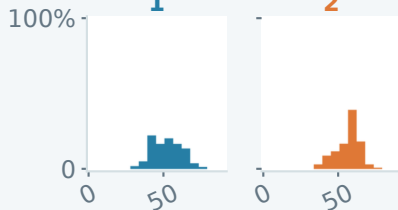


Age (years)

Histogram 96.69

1

2



+15.65 points | best of 7 familiar charts

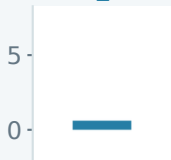
c Exercise-related ECG change

How do five group gaps differ?

1 No disease

2 Disease

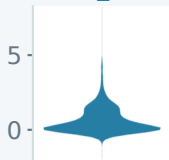
Bars 64.22
1



2



Violin 92.48
1



2



ST depression (source units)

+28.26 points | best of 7 familiar charts

d Age in the heart-failure cohort

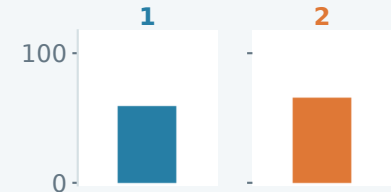
How do five group gaps differ?

1 Survived

2 Died

Bars 79.38

Heatmap 96.27



Age (years)

+16.89 points | best of 7 familiar charts

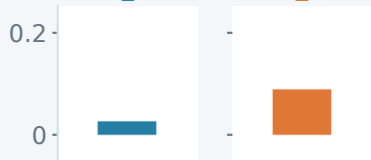
e Cell-nucleus boundary shape

How do the spreads differ?

1 Benign

2 Malignant

Bars 53.74
1



Box plot 98.65
1



Concave points (source units)

+44.91 points | best of 7 familiar charts

f Heart-failure cohort composition

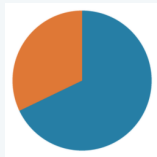
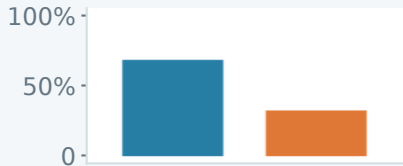
What share is in each category?

1 Survived

2 Died

Bars 99.725

Pie 99.950



Observed record shares; not population risk

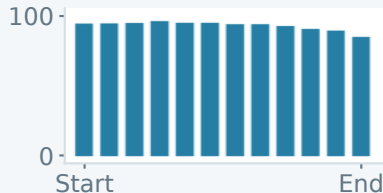
+0.225 points | best of 4 familiar charts

g Heart rate over one recording

How does heart rate change over time?

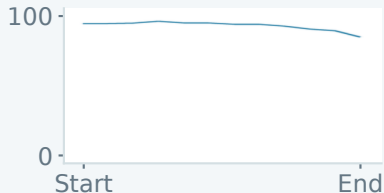
BIDMC recording 04

Bars 90.22



12 successive time bins; beats/min

Line 96.84



+6.62 points | best of 4 familiar charts

h ECG-pulse association

How are ECG and pulse rates related?

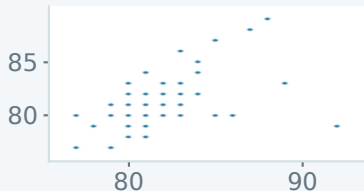
Paired observations from BIDMC 22

Heatmap 99.206



Horizontal: ECG; vertical: pulse (beats/min)

Scatter 99.803



+0.597 points | best of 2 familiar charts

a Acceleration-change variability

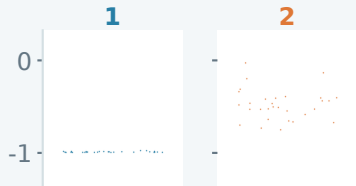
How do five group gaps differ?

1 Sitting

2 Walking

Dots 82.35

Heatmap 84.09



Normalized source units

Choice changes | gap 1.74 points

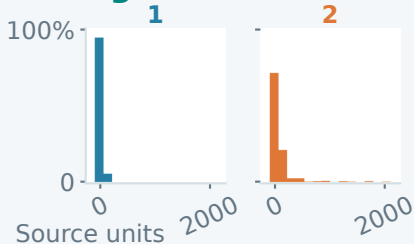
b Liver enzyme ALT

How do five group gaps differ?

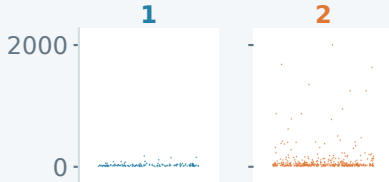
1 No liver disease

2 Liver disease

Histogram 95.12



Dots 96.66



Choice changes | gap 1.55 points

c Gravity-signal variability

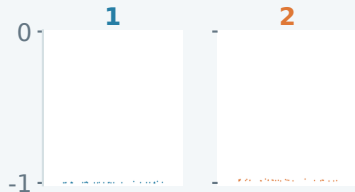
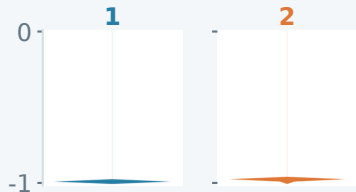
How do five group gaps differ?

1 Sitting

2 Walking

Violin 75.95

Dots 81.12



Normalized source units

Choice changes | gap 5.16 points

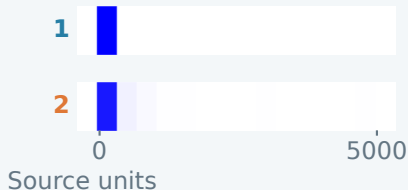
d Liver enzyme AST

How do five group gaps differ?

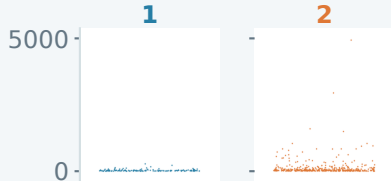
1 No liver disease

2 Liver disease

Heatmap 88.73



Dots 90.64



Choice changes | gap 1.90 points

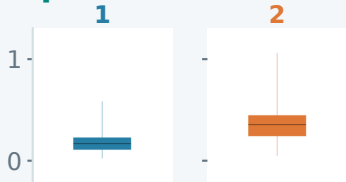
e Cell-nucleus compactness

How do the spreads differ?

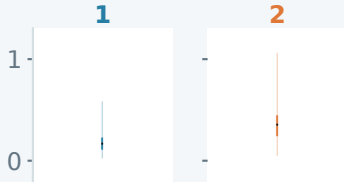
1 Benign

2 Malignant

Box plot 98.02



Interval 99.59



Compactness (source units)

Choice changes | gap 1.57 points

f Kidney-cohort composition

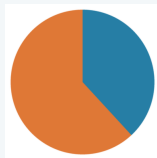
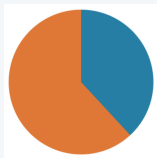
What share is in each category?

1 Non-CKD

2 CKD

Pie 99.943

Pie 99.943



Observed record shares

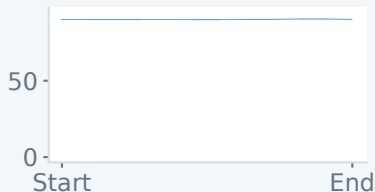
Choice retained

g Heart rate over one recording

How does heart rate change over time?

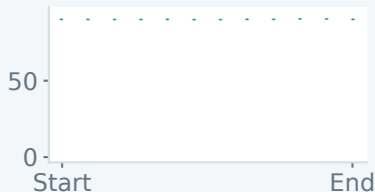
BIDMC recording 07

Line 61.66



12 time bins; beats/min

Dots 68.14



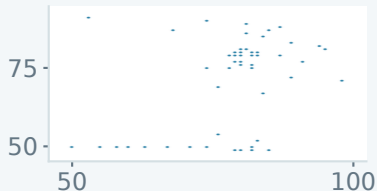
Choice changes | gap 6.48 points

h ECG-pulse association

How are ECG and pulse rates related?

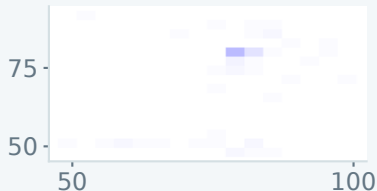
Paired observations from BIDMC 41

Scatter 98.884



Horizontal: ECG; vertical: pulse (beats/min)

Heatmap 99.170



Choice changes | gap 0.286 points